TASK 2 / DATA SCIENCE RESEARCH BRIEF

Unemployment in India

The 2020 lockdown and labour-market rebound

Monthly evidence from regional and urban/rural labour-market estimates.

May 2019 to October 2020 across two separate source series.

FIXED REGIONAL PANEL	JAN-FEB	APR-MAY	JUN-OCT
Mean unemployment	8.76%	23.20%	9.52%

Each figure is an unweighted regional mean.

Executive summary

The unweighted unemployment mean across 25 matched regions rose from 8.76% in January-February to 23.20% in April-May (+14.44 percentage points, +164.8% relative). The monthly mean peaked in May at 23.37%. Across April-October, the mean was 13.43% (+4.67 pp; +53.3% relative to baseline).

The June-October mean fell to 9.52% (+0.76 pp from baseline), reaching 7.99% in October. In that final month participation remained 2.89 pp below baseline and mean estimated employment was 2.7% lower.

Haryana had the highest observed-month mean (27.48%), Meghalaya the lowest (3.87%), and Puducherry the largest within-region SD (26.59 pp). The largest paired increases were Puducherry (+65.85 pp), Jharkhand (+41.93 pp), Tamil Nadu (+39.66 pp). The highest individual estimates were 75.85% in Puducherry (April 2020, region-wide) and 76.74% in Puducherry (April 2020, urban).

Across 20 regions with both areas, urban unemployment rose from 12.20% to 27.92% (+15.72 pp); rural unemployment rose from 9.00% to 23.31% (+14.31 pp). The estimated urban-minus-rural change was +1.41 pp (95% region bootstrap interval -2.42 to 5.14 pp). The interval spans zero, leaving the difference uncertain.

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Sources and study design

The study measures unemployment variation, quantifies changes around the lockdown, examines local and urban/rural differences, and evaluates the evidence for recurring monthly patterns.

Acquisition and coverage

SOURCE GRAIN	RAW ROWS	VALID ROWS	REGIONS	OBSERVED MONTHS
Region x area	768	740	28	May 2019 to Jun 2020
Region-wide	267	267	27	Jan 2020 to Oct 2020

Source files were retrieved via the official Kaggle ZIP endpoint on 2026-11-09 (UTC date). The manifest records file sizes and SHA-256 hashes. The region-wide series ends in October 2020.

Quality and comparability

Removed 28 fully blank records. The valid rows had no missing fields, duplicate observation keys or out-of-bounds labour values. All 45 IQR flags were retained. Dates were parsed day-first, frequencies standardized and the transposed coordinate headers corrected.

The files overlap in 158 region-month keys, with both areas present in 148. In 144 of those, the regional rate differs from the simple area mean by more than 0.011 pp. Area percentages require labour-force weights to aggregate. The two files serve separate analytical roles.

Coverage audits show 44 absent region-month-area combinations and 3 absent region-month combinations. Missing cells were left unfilled.

Period and aggregation rules

January-February defines baseline. March is a transition month because the nationwide lockdown took effect on 25 March [2]. April-May captures the first full lockdown months. Reopening starts in June [3] and ends at the last available month (October for region-wide data, June for area data). The pandemic average pools observed April-onward months equally.

Comparisons fix the set of regions and weight regions and months equally. The area comparison uses the same regions in both groups. Unemployment is unemployed persons as a share of the labour force. Percentage-point change is current minus baseline. Relative change divides that difference by baseline and multiplies by 100. Employment figures are mean counts per region or area.

Distribution and geography

Across all 267 region-month estimates, unemployment averaged 12.24% with a median of 9.65% and SD of 10.80 pp. Values ranged from 0.50% to 75.85%, with the middle half between 4.84% and 16.75% (IQR 11.91 pp).

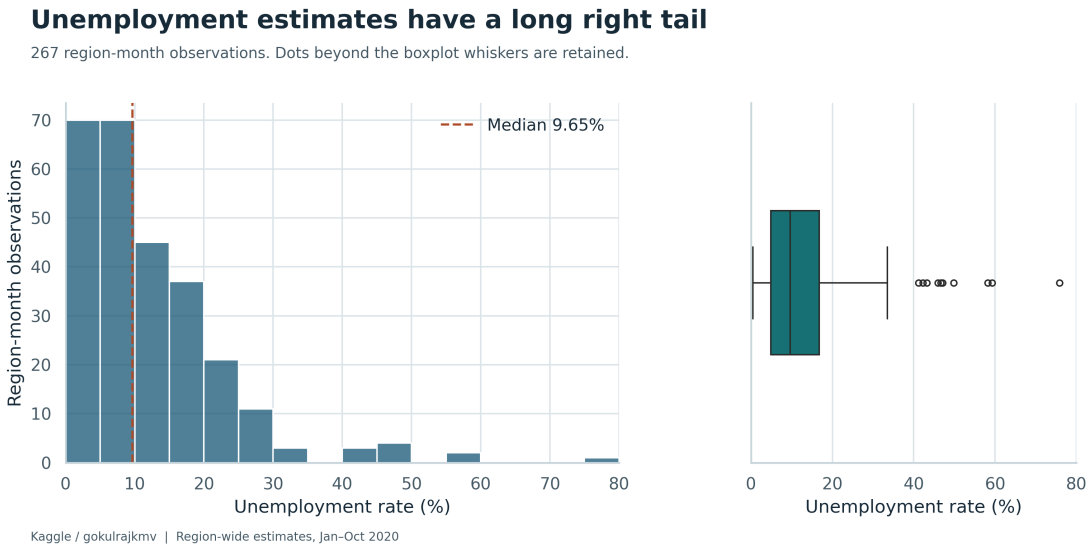


Figure 1. Mean unemployment exceeds the median because a small share of observations are very high.

Haryana had the highest observed-month mean (27.48%), Meghalaya the lowest (3.87%), and Puducherry the largest within-region SD (26.59 pp). The largest paired increases were Puducherry (+65.85 pp), Jharkhand (+41.93 pp), Tamil Nadu (+39.66 pp). The highest individual estimates were 75.85% in Puducherry (April 2020, region-wide) and 76.74% in Puducherry (April 2020, urban).

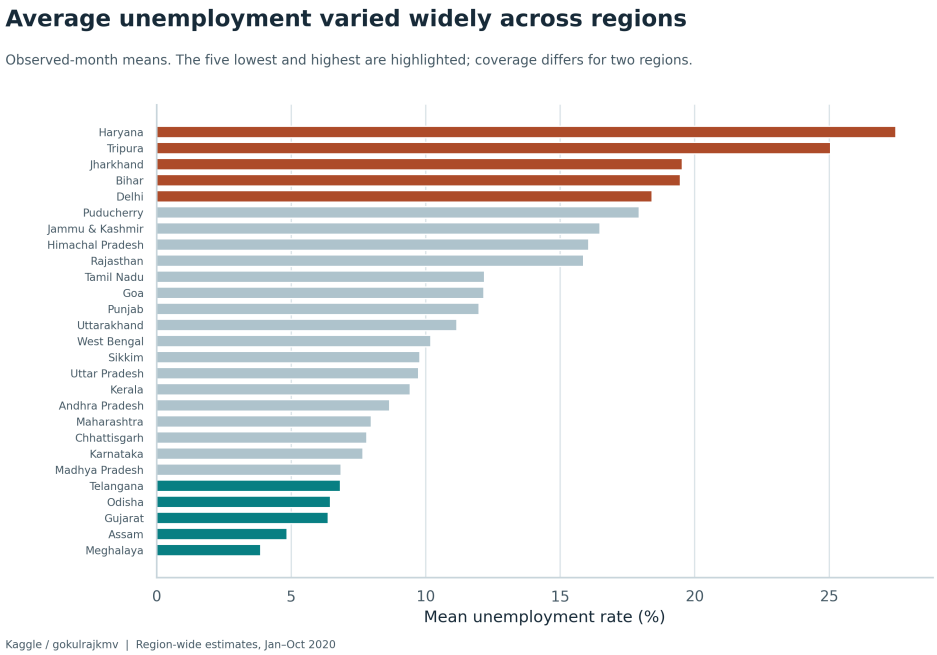


Figure 2. Ranks use every available month in each region. Two regions have incomplete coverage.

Lockdown and recovery

The unweighted unemployment mean across 25 matched regions rose from 8.76% in January-February to 23.20% in April-May (+14.44 percentage points, +164.8% relative). The monthly mean peaked in May at 23.37%. Across April-October, the mean was 13.43% (+4.67 pp; +53.3% relative to baseline).

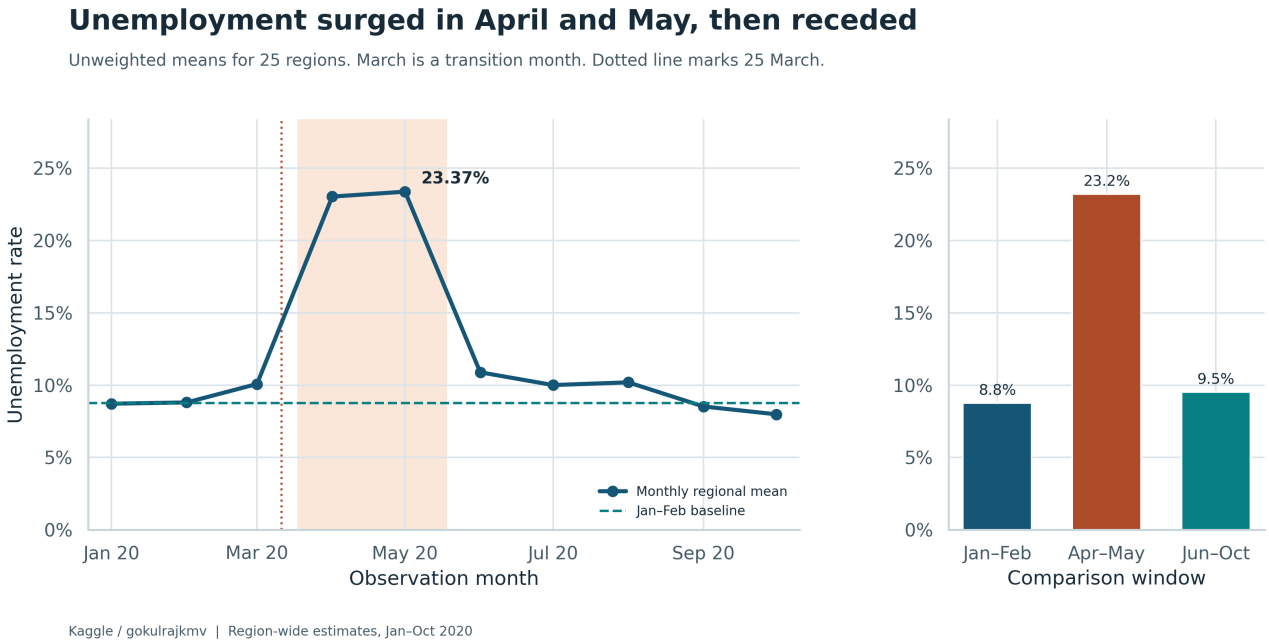


Figure 3. The fixed-panel peak was in May, followed by a steep decline in June.

The June-October mean fell to 9.52% (+0.76 pp from baseline), reaching 7.99% in October. In that final month participation remained 2.89 pp below baseline and mean estimated employment was 2.7% lower.

During April-May, mean participation fell from 44.55% to 37.33% (-7.22 pp). Mean estimated employment fell from 16.05 million to 11.64 million per region (-27.5%).

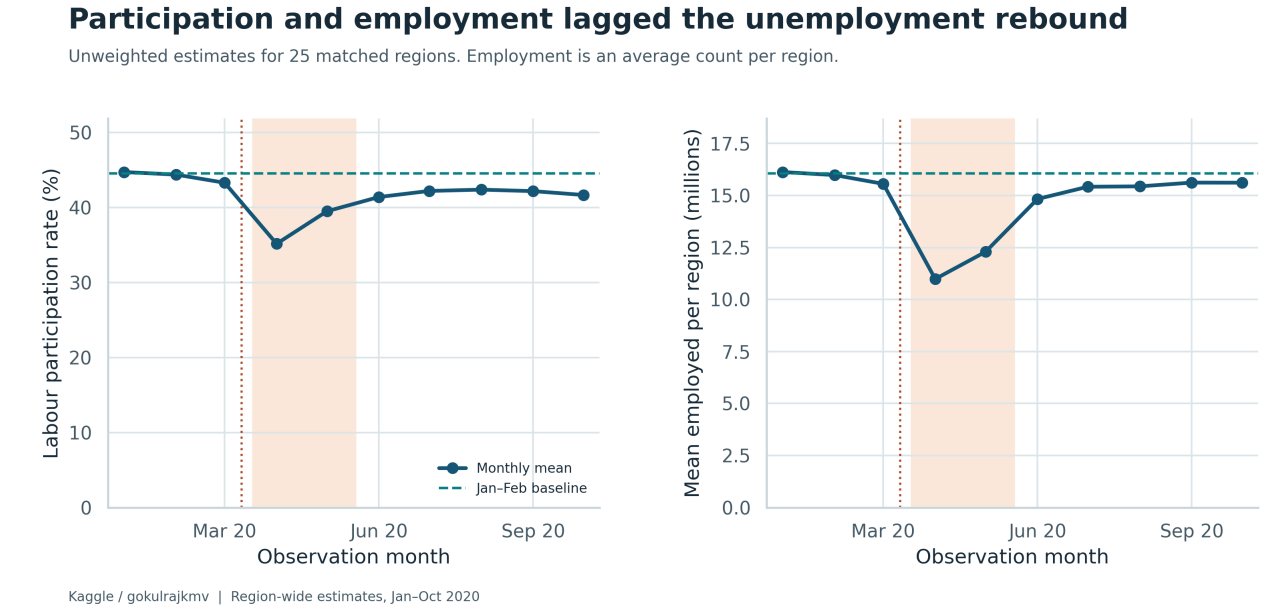


Figure 4. The two companion measures remained below their January-February averages in October.

Regional and area contrasts

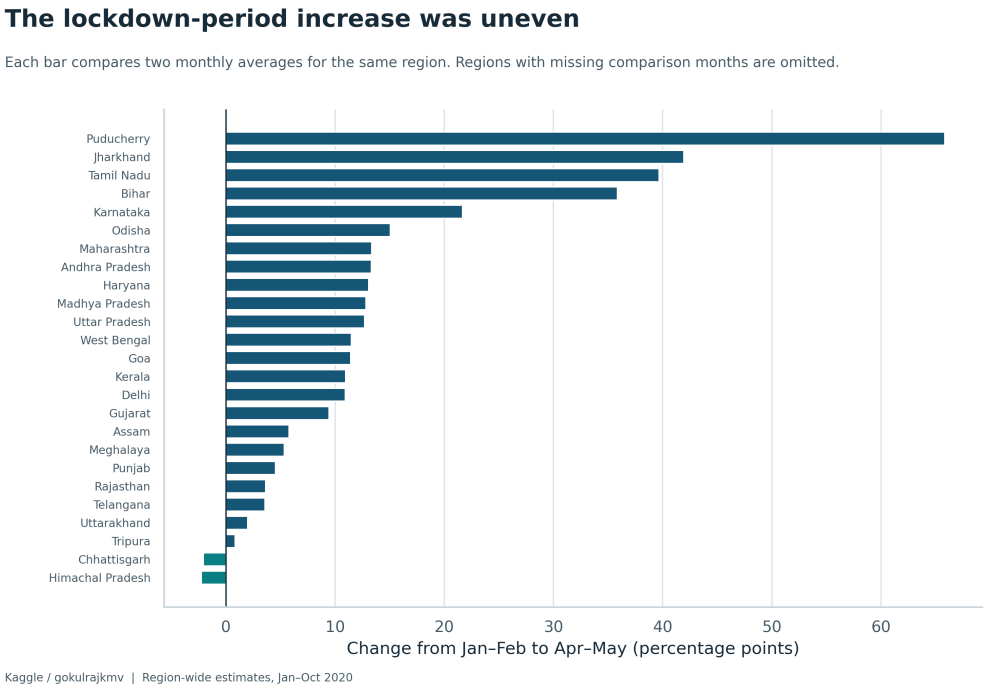


Figure 5. Within-region changes span declines and increases above 40 percentage points.

Across 20 regions with both areas, urban unemployment rose from 12.20% to 27.92% (+15.72 pp); rural unemployment rose from 9.00% to 23.31% (+14.31 pp). The estimated urban-minus-rural change was +1.41 pp (95% region bootstrap interval -2.42 to 5.14 pp). The interval spans zero, leaving the difference uncertain.

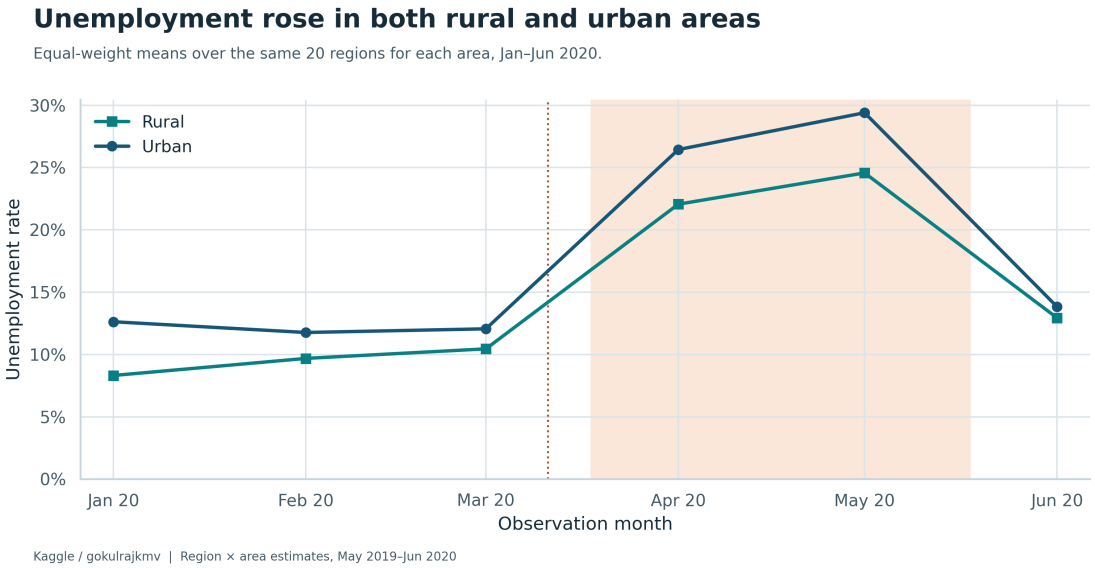


Figure 6. Both areas showed a sharp April-May increase and a lower June reading.

Inference and associations

Unemployment rose in 23 of 25 regions (median change +11.43 pp). The bootstrap 95% interval for the mean change was 8.92 to 21.09 pp. The two-sided sign test gave p = 1.9e-05; paired Cohen's dz was 0.91. Shapiro-Wilk p = 0.00015 suggests asymmetric or non-normal differences. Leaving out one region at a time put the mean change between 12.30 and 15.14 pp.

The resampling unit is the region, using 20,000 draws of paired changes. The sign test assesses balance of positive and negative changes, avoiding a normality assumption. The national shock can correlate regional responses, so both inference summaries are exploratory and conditional on the observed regions.

Restricting completeness to January-May gives shock changes of +14.44 pp versus +14.44 pp in the full-window panel.

The pooled Pearson correlations of unemployment with estimated employment and participation were -0.25 and -0.07. After removing each region's average, they were -0.58 and -0.06. The within-region associations can still reflect common shocks and linked labour-market definitions.

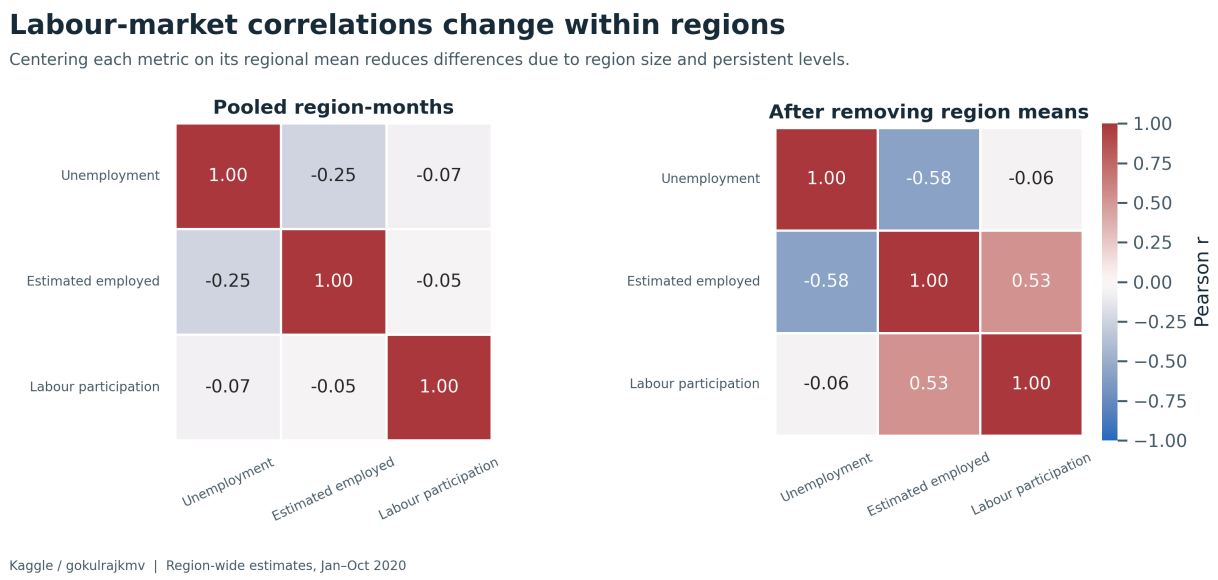


Figure 7. Regional demeaning changes the magnitude of the employment and participation relationships.

Interpretation limits

The regional panel observes the same broad shock at the same time. The short pre-period, lack of a comparison group and unknown sampling weights limit interpretation. These results describe associations with the lockdown period. They cannot isolate effects of lockdown, health risk, policy responses, concurrent changes or changes in survey conditions.

Time patterns and seasonality

Only May and June repeat across the 14-month area series. The matched year-over-year differences were +17.60 pp in May and +3.53 pp in June. The largest month-to-month increase was +13.00 pp in April 2020. With fewer than two annual cycles and pandemic disruption in one of the years, these comparisons are exploratory. STL and forecasting were omitted.

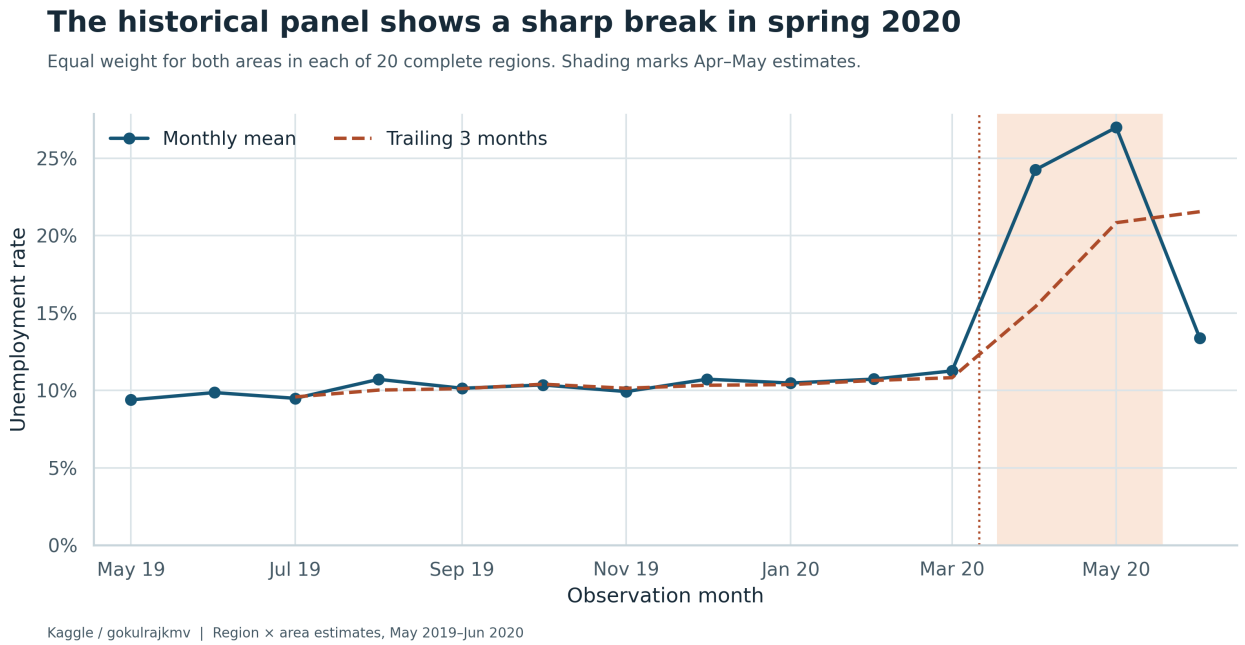


Figure 8. The trailing three-month mean smooths the spike and carries it into June.

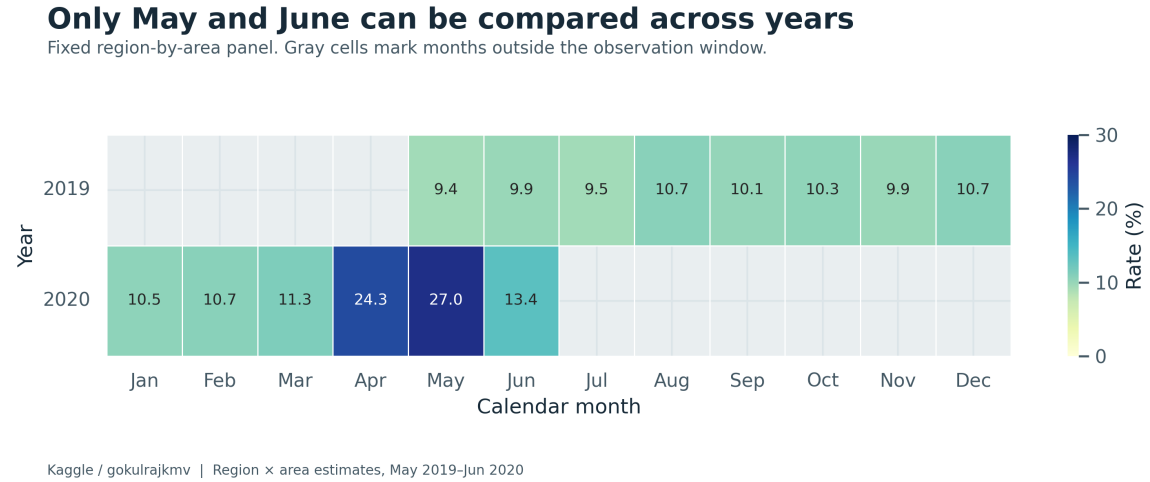


Figure 9. Unobserved months remain blank. The highest calendar averages are driven by 2020.

Partial-year averages were 10.07% across 8 months in 2019 and 16.18% across 6 months in 2020. The two averages cover different calendar months.

Policy implications

Target regional response

Pilot rapid-response placement and payroll support where unemployment increased most, beginning with Puducherry and other severely affected regions. Validate local indicators and employer demand before allocating support, then track re-employment and retention.

Adapt support by area

Test job-matching and temporary income support for urban workers alongside rural public works for households facing a sudden loss of work. Both area means rose sharply, and the evidence leaves the urban-rural difference uncertain. Track access and outcomes separately by area.

Track who returns to work

Include participation and employment alongside unemployment in recovery dashboards. October unemployment had fallen below baseline while the other two measures remained lower. Use survey weights and worker-level follow-up to identify groups that stopped looking for work.

Conclusion

Unemployment rose sharply in the first full lockdown months and fell during phased reopening. Large local differences and continued shortfalls in participation and employment make the unemployment rate alone an incomplete measure of recovery. The panel ends in 2020 and cannot show durable recovery.

References

[1] Gokulraj K M V. Unemployment in India. Kaggle dataset release.

[2] Government of India, Press Information Bureau. Nationwide lockdown order announcement. 24 March 2020.

[3] Government of India, Press Information Bureau. Guidelines effective 1 June 2020. 30 May 2020.